Stavros X. Drakopoulos

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EDUCATION

Loughborough University — Loughborough, UK
PhD in Materials Engineering — Oct 2016 – Sept 2019
PhD Thesis Title: Disentangled ultra-high molecular weight polyethylene and its nanocomposites: Relaxation dynamics, entanglement formation and anisotropic properties due to orientation
Advisors: Sara Ronca, Ignacio Martin-Fabiani, Simon Martin

University of Patras — Patras, Greece
BSc in Materials Science — Oct 2011 – Aug 2016
Undergraduate Thesis Title: Chemical composition, thermomechanical behavior and electrical response of biocomposite thermoplastic starch matrix materials, reinforced with latex and cellulose particles
Advisor: Georgios C. Psarras

PROFESSIONAL EXPERIENCE

Princeton University — Princeton, NJ, USA
Associate Research Scholar — 2023 – present
Advisor: Rodney D. Priestley
Research Topic: *Relaxation and Charge Transport Dynamics of Polymer-based Systems*

University of Bath — Bath, UK
Postdoctoral Research Associate — 2021 – 2023
Advisor: Kamal Asadi
Research Topic: *Dielectric Spectroscopy of Conjugated Polymers, Polyelectrolytes, and Polymer-Grafted Nanoparticles*

University of Birmingham — Birmingham, UK
Faraday Institution Research Fellow — 2020 – 2021
Advisor: Emma Kendrick
Research Topic: *Electrode Manufacturing for Li-ion Batteries*

AWARDS AND HONORS

PMSE Future Faculty Scholars — 2026
American Chemical Society, Division of Polymeric Materials: Science & Engineering.

Peter Debye Prize for Young Investigators for Excellence in Dielectric Research — 2024
Biannually awarded by the International Dielectric Society

D.H. Richards Memorial Travel Bursary — 2019
Macro Group, Royal Society of Chemistry.

Studentship for Doctoral Research — 2016 – 2019
Loughborough University

Erasmus Placement Scholarship — 2014
University of Patras, Greece.

PUBLICATIONS

**denotes corresponding author(s) / [§]denotes equal contribution*

Drafts and Submissions

32. **S.X. Drakopoulos**, J. Guo, X. Wu, O. Vryonis, R.D. Priestley*, L.S. Xie* (2027) 2D metal–organic chalcogenides/polymer nanodielectrics for capacitive energy storage. Under preparation.
31. **S.X. Drakopoulos**, R.D. Priestley* (2027) Self-healing polymer dielectrics for capacitive energy storage applications: A review. To be submitted soon.
30. **S.X. Drakopoulos**[§], Y. Seo[§], R.D. Priestley* (2027) Influence of ageing and water content on the relaxation dynamics of hydroxypropyl methylcellulose acetate succinate. To be submitted soon.
29. **S.X. Drakopoulos**[§], M.M. Abolhasani[§], K. Asadi* (2026) Dielectric relaxation and charge transport dynamics in polymer grafted metallic nanoparticles. Under Review in *Nanoscale Advances* (RSC).

Journal Publications

28. **S.X. Drakopoulos***, S. Kim, R.A. Register, R.D. Priestley* (2026) Segmental dynamics and local motions in disordered random copolymers. *Macromolecules* 59(8):4787-4799.
27. H.-L. Sun[§], **S.X. Drakopoulos**[§], L. Čamdžić, S. Maguire, R.D. Priestley*, E.E. Stache* (2026) Polyacrylate vitrimer network via in-situ isocyanide copolymerization: Synthesis and molecular dynamics. *Journal of the American Chemical Society* 148(10):11047-11057.
26. **S.X. Drakopoulos*** (2025) Revisiting the dielectric spectrum: Tricks and treats of analysis and interpretation around the conductivity relaxation. *IET Nanodielectrics* 8(1):e70020.
25. S.M. Maguire, C. Nie, S.V. Sundar, **S.X. Drakopoulos**, P.J. Chirik, R.D. Priestley*, E.C. Davidson* (2025) Modulating poly(oligocyclobutane) properties through backbone modifications. *Macromolecules* 58(17):9469-9482.
24. **S.X. Drakopoulos***[§], K. Loukelis[§], M. Triantafyllou-Rundell, C.C. Stoumpos, M. Chatzinikolaidou, G.C. Psarras (2025) Fabrication of all-organic nanodielectrics reinforced with electrospun polymer fibres for capacitive energy storage. *RSC Applied Polymers* 3:960-972.
23. **S.X. Drakopoulos***, O. Vryonis, Z. Špitalský, H. Peidayesh, L. Lendvai* (2024) Thermoplastic starch processed under various manufacturing conditions: thermal and electrical properties. *Biomacromolecules* 25(9):5938–5948.
22. **S.X. Drakopoulos**, J. Wu, S.M. Maguire, S. Srinivasan, K. Randazzo, E.C. Davidson, R.D. Priestley* (2024) Polymer nanocomposites: Interfacial properties and capacitive energy storage. *Progress in Polymer Science* 156:101870.
21. **S.X. Drakopoulos***, K. Loukelis, M. Triantafyllou-Rundell, C.C. Stoumpos, M. Chatzinikolaidou, G.C. Psarras (2024) Epoxy/clay nanodielectrics: From relaxation dynamics to capacitive energy storage. *Advanced Composites and Hybrid Materials* 7:118.
20. **S.X. Drakopoulos***, J. Cui, M. Asandulesa, P.W.M. Blom, A. Nogales, K. Asadi* (2024) Universal scaling of DC conductivity with dielectric interfacial polarization in conjugated polymers. *Macromolecules* 57(6):2661-2668.
19. **S.X. Drakopoulos***, Z. Špitalský, H. Peidayesh, L. Lendvai* (2023) The effect of drying of glycerol-plasticized starch upon its dielectric relaxation dynamics and charge transport. *Journal of Polymers and the Environment* 31:5389-5400.
18. **S.X. Drakopoulos**, T. Cowell, E. Kendrick* (2023) Graphite-SiO_x electrodes with a biopolymeric binder for Li-ion batteries: Predicting the cycle life performance from physical properties. *ACS Applied Energy Materials* 6(12):6543-6553.
17. **S.X. Drakopoulos***, J. Yang, O. Vryonis, L. Williams, G.C. Psarras, E. Mele (2022) Flexible polymer-based nanodielectrics reinforced with electrospun composite nanofibres for capacitive energy storage. *ACS Applied Polymer Materials* 4(11):8203-8215.
16. T. Kim, **S.X. Drakopoulos**, S. Ronca, A.J. Minnich* (2022) Origin of high thermal conductivity in disentangled ultra-high molecular weight polyethylene films: ballistic phonons within enlarged crystals. *Nature Communications* 13:2452.
15. **S.X. Drakopoulos***, A.C. Patsidis, G.C. Psarras (2022) Epoxy-based $BaTiO_3$ nanodielectrics: Relaxation dynamics, charge transport and energy storage. *Materials Research Bulletin* 145:111537.
14. **S.X. Drakopoulos**, A. Gholamipour-Shirazi, P. MacDonald, R.C. Parini, C.D. Reynolds, D.L. Burnett, B. Pye, K.B. O'Regan, G. Wang, T.M. Whitehead, G.J. Conduit, A. Cazacu, E. Kendrick* (2021) Formulation and manufacturing optimization of lithium-ion graphite-based electrodes via machine learning. *Cell Reports Physical Science* 2:100683.

13. **S.X. Drakopoulos***, G.C. Manika, A. Nogales, T. Kim, A.B. Robbins, G. Claudio, A.J. Minnich, T.A. Ezquerra, G.C. Psarras I. Martin-Fabiani, S. Ronca (2021) Gold/ultra-high molecular weight polyethylene nanocomposites for electrical energy storage: Enhanced recovery efficiency upon uniaxial deformation. *Journal of Applied Polymer Science* 138:51232.
12. I. Martin-Fabiani*, **S.X. Drakopoulos**, G. Forte, S. Prévost, I. Hoffmann, S. Ronca (2021) Chain dynamics of ultra-high molecular weight polyethylene composites with graphene oxide nanosheets. *ACS Macro Letters* 10:460-465.
11. **S.X. Drakopoulos***, G.C. Psarras S. Ronca (2021) Oriented ultra-high molecular weight polyethylene/gold nanocomposites: Electrical conductivity and chain entanglement dynamics. *eXPRESS Polymer Letters* 15(6):492-502.
10. **S.X. Drakopoulos***, O. Tarallo, L. Guan, I. Martin-Fabiani, S. Ronca (2020) Nanocomposites of Au/disentangled UHMWPE: A combined optical and structural study. *Molecules* 25:3225.
9. **S.X. Drakopoulos***, J. Karger-Kocsis, G.C. Psarras (2020) The effect of micro-fibrillated cellulose upon the dielectric relaxations and DC conductivity in thermoplastic starch bio-composites. *Journal of Applied Polymer Science* 137(48):49573.
8. **S.X. Drakopoulos***, G. Forte, S. Ronca (2020) Relaxation dynamics in disentangled ultrahigh molecular weight polyethylene via torsional rheology. *Industrial & Engineering Chemistry Research* 59:4515-4523.
7. A.B. Robbins, **S.X. Drakopoulos**, I. Martin-Fabiani, S. Ronca, and A.J. Minnich* (2019) Ballistic thermal phonons traversing nanocrystalline domains in oriented polyethylene. *Proceedings of the National Academy of Sciences* 116:17163-17168.
6. **S.X. Drakopoulos**, J. Karger-Kocsis, G.C. Psarras* (2019) In situ thermodielectric analysis of the gelatinization mechanism of raw maize starch: An experimental and theoretical approach. *Journal of Polymers and the Environment* 27:333-342.
5. **S.X. Drakopoulos***, G.C. Psarras, G. Forte, I. Martin-Fabiani, S. Ronca (2018) Entanglement dynamics in ultra-high molecular weight polyethylene as revealed by dielectric spectroscopy. *Polymer* 150:35-43.
4. D. Nocita*, G. Forte, **S.X. Drakopoulos**, A. Visco, A. Gianporcaro, S. Ronca (2017) Processing and characterization of bio-polyester reactive blends: From thermoplastic blends to cross-linked networks. *Polymer* 132:256-263.
3. **S.X. Drakopoulos**, J. Karger-Kocsis, Á. Kmetty, L. Lendvai, G.C. Psarras* (2017) Thermoplastic starch modified with microfibrillated cellulose and natural rubber latex: A broadband dielectric spectroscopy study. *Carbohydrate Polymers* 157:711-718.
2. L. Lendvai, J. Karger-Kocsis*, Á. Kmetty, **S.X. Drakopoulos** (2016) Production and characterization of microfibrillated cellulose reinforced thermoplastic starch composites. *Journal of Applied Polymer Science* 133:42397.
1. J. Karger-Kocsis, Á. Kmetty, L. Lendvai, **S.X. Drakopoulos**, T. Bárány* (2015) Water-assisted production of thermoplastic nanocomposites: A review. *Materials* 8:72-95.

Book Chapters

2. **S.X. Drakopoulos** (2024) Dielectric Relaxation and Transport Dynamics of Solid-State Polymer Electrolytes. In '*Batteries – The Future of Energy Storage*' (Ed: S. Maniam) published by Jenny Stanford Publishing, pp. 117-153.
1. **S.X. Drakopoulos**, S. Ronca, I. Martin-Fabiani (2020) Dielectric behavior of nonpolar polymers and their composites: The case of semicrystalline polyolefins. In '*Dielectrics and Crystallization*' (Eds: T.A. Ezquerra & A. Nogales) part of the *Advances in Dielectrics* book series published by Springer International Publishing, pp. 243-265.

Patents

2. **S.X. Drakopoulos** (November 2016) A Method of Improving the Piezoelectric Charge Coefficient in Composite Piezoelectric Materials. *Hellenic Industrial Property Organization* GR 1009017.
1. **S.X. Drakopoulos** (December 2014) Piezoelectric Engine. *Hellenic Industrial Property Organization* GR 1008662.

INVITED PRESENTATIONS

6. Seminar on segmental dynamics in random co-polymers to the Computation and Theoretical Society at the Beckman Institute at Illinois, University of Illinois Urbana-Champaign, Illinois, USA.
5. Presentation on Segmental dynamics and local motions in disordered random co-polymers as an Awardee in the PMSE Future Faculty Symposium at the ACS Fall 2026 in Chicago, Illinois, USA.
4. Presentation on Dielectric properties and relaxation dynamics of poly(acrylate-co-isocyanide) vitrimers at the 10th International Discussion Meeting on Relaxations in Complex Systems 2025, in Barcelona, Spain.
3. Departmental Seminar and Career Lecture to undergraduate students, graduate students, and faculty, Department of Materials Science, University of Patras, Greece, 2023.
2. Seminar on disentangled UHMWPE and entanglement formation dynamics to researchers and faculty, Department of Biobased Materials, Maastricht University, The Netherlands, 2018.
1. Departmental Seminar and Career Lecture to undergraduate students, graduate students, and faculty, Department of Materials Science, University of Patras, Greece, 2017.

PRESENTATIONS IN INTERNATIONAL CONFERENCES

18. AIChE Annual Meeting (2026) Minneapolis, MA, USA
17. ACS Fall (2026) Chicago, Il, USA
16. Polymer Physics Gordon Research Conference (2026) South Hadley, MA, USA
15. International Discussion Meeting on Relaxations in Complex Systems (2025) Barcelona, Spain
14. APS Joint March Meeting and April Meeting: Global Physics Summit (2025) Anaheim, CA, USA
13. Physical Aspects of Polymer Science (2024) Edinburg, UK
12. 12th Conference on Broadband Dielectric Spectroscopy and its Applications (2024) Lisbon, Portugal
11. International Conference on Science and Technology of Synthetic Electronic Materials (2024) Dresden, Germany
10. APS March Meeting (2024) Minneapolis, Minnesota, USA
9. Physical Aspects of Polymer Science (2022) London, UK
8. APS March Meeting (2020) Denver, Colorado, USA
7. 5th Blue Sky Conference on Catalytic Olefin Polymerization (2019) Naples & Sorrento, Italy
6. 6th International Symposium Frontiers in Polymer Science (2019) Budapest, Hungary
5. 29th European Conference on Biomaterials (2018) Maastricht, Netherlands
4. 10th Conference on Broadband Dielectric Spectroscopy and its Applications (2018) Brussels, Belgium
3. Physical Aspects of Polymer Science (2017) Swansea, UK
2. 1st International Conference 'Science in Technology' (2015) Athens, Greece
1. 10th Hellenic Polymer Society Conference, with International Participation (2014) Patras, Greece

TEACHING EXPERIENCE

Princeton University
EGR152, Foundations of Engineering: The Mathematics of Shape and Motion, Lecturer — Fall 2026

Rowan University
CHE06502, Special Topics in CHE: Polymer-1, Guest Lecturer — Spring 2026

Loughborough University
MPP501, Polymer Science, Guest Lecturer — Spring 2018 & 2019
CGA014, Mechanical Properties of Polymers, Teaching Assistant — Fall 2018

MENTORING EXPERIENCE

Princeton University

Kaitlyn Zuravel, Undergraduate Student (currently Graduate Student at Rice University, TX) — Spring 2024
Outcome: Kaitlyn presented our work at *AIChE 2024*. — Summer 2024

University of Birmingham

Thomas Cowell, Undergraduate Student — Summer 2021
Co-authored *ACS Applied Energy Materials*

Loughborough University

Jing Yang, Master's Student (later obtained PhD from the University of Manchester, UK) — Summer 2019
Co-authored *ACS Applied Polymer Materials*

Linlin Guan, Master's Student (later obtained MPhil from the University of Manchester, UK) — Summer 2018
Co-authored *Molecules*

Jing Yang, Undergraduate Student — Spring 2018

OUTREACH

Interview at UP Magazine (in *Greek*) @up (2025) 21:19-21. — 2025
This is the official magazine of the University of Patras, Greece.

Non-scientific article to promote materials science (in *Greek*) — 2017
MENSA Magazine (2017) 42:46-47.

Invited presentation to high school students to promote materials science — 2015
4th High School of Petroupolis, Athens, Greece

PROFESSIONAL ACTIVITIES – SERVICE

Journal Review

Nature Communications, Newton, Journal of the American Chemical Society, ACS Applied Materials & Interfaces, ACS Applied Electronic Materials, ACS Applied Polymer Materials, Carbohydrate Polymers, Crystals, Industrial & Engineering Chemical Research, Industrial Crops and Products, JACS Au, Journal of Materials Science, Journal of Materials Science and Technology, Journal of Polymers and the Environment, Macromolecules, Micro & Nano Letters, npj Computational Materials, Polymer Bulletin, Polymer Engineering & Science, RSC Advances, Soft Matter, New Journal of Chemistry, Langmuir, Physical Chemistry Chemical Physics, Electrical Engineering, Science China Chemistry, Advanced Composites and Hybrid Materials.

Society Memberships

American Institute of Chemical Engineering (AIChE) — 2026 – present
American Chemical Society (ACS) — 2026 – present
American Physical Society (APS) — 2025 – present
Royal Society of Chemistry (RSC) — 2018 – present
International Dielectric Society (IDS) — 2018 – present